

COPILOTTRAINING
TECHNICAL BRIEFING

Copilot on GitHub

Produce, Reason, Measure

Produce, reason, measure. Then a person decides.

Agents on GitHub · CopilotTraining Tech Talk · 50 min

🤔 THE CORE QUESTION

"Which built-in GitHub agent fits each PR moment, and what evidence does it hand the decider?"

The cloud agent produces, code review reasons, and Code Quality measures.

Then a person decides.

Requester and PR owner

Delegate the issue, then answer findings and update the head

Reviewer

Validate behavior and accept the residual risk at merge

Platform owner

Own invocation, permissions, required evidence, and rollback

12 seconds

Assignment of #7 to draft #210 opening (observed, one FanHub run)

8 of 8

Tests passed when the maintainer ran what the review left unexecuted

26.7% → 30%

A coverage rule on #199 blocked the first head, then passed

TABLE OF CONTENTS

PART 1

Delegate the Change

Issue #7 to a tested draft in under four minutes

Hand off a scoped issue and run the PR-owner acceptance check

PART 2

Reason About Intent

Which head review saw and what stayed unexecuted

Read the analyzed commit, run the named tests, weigh preview approvals

PART 3

Measure Against Rules

Findings, ratings, and coverage thresholds

Use rules for what rules measure and start every gate in Evaluate

PART 4

Decide What Gets Authority

Permit, count, enforce: one promotion at a time

Map evidence to roles, promote one setting, and write the decision

Four Kinds of Evidence on #210's Final Head

Commit 02ddaf3 of FanHub draft #210, at the moment a person decides (observed)

Patch + tests	Two-file fix to the episode cache with eight new regression tests	`cloud agent → PR owner, reviewer`
Assessment	"🟢 Approval recommended … tests were not executed"	`code review → reviewer`
Measurement	A Code Quality run on 02ddaf3 with no findings	`Code Quality → reviewer, platform`
Executed run	The maintainer ran the tests: 8 of 8 passed	`maintainer → reviewer`

Each row answers a different question for requester, PR owner, reviewer, and platform owner

PART 1

Delegate the Change

Assign issue #7 and follow the cloud agent from a plan commit to a tested fix on draft #210

⏱️

12 s to a draft

Observed: assigned 20:31:19, draft opened 20:31:31

✅

Acceptance check

Five questions before the PR owner reads the diff

🔒

Platform-fixed

The agent drafts; a person marks ready and merges

```
maintainer assigns FanHub #7 to Copilot
↳ draft #210 → fix 02ddaf3 + 8 tests in under 4 min (observed)
```

One Assignment, One Draft, Under Four Minutes

Issue #7: the Node frontend episode cache returns wrong episodes after a season filter

```
$ FanHub issue #7 · 20:31:19 · maintainer
> "Assign to Copilot"
Copilot cloud agent (ephemeral Actions environment):
  20:31:31 · opens draft #210 with an Initial plan commit (919b158)
  20:34:55 · pushes 02ddaf3: Episodes.js caches only unfiltered responses and ignores stale requests
  Adds Episodes.test.js: 8 regression tests for filters, expiry, remounts, and races
  PR body: Fixes #7 · 2 files · +203 / -7
✅ The agent produced the draft, branch, and tests. The merge stays with a person.
```

issue #7 → draft (plan) → fix + 8 tests → review → quality → maintainer decides · one observed run

Delegate Scoped Work, Then Check the Head Before the Diff

🧭 What the requester delegates

Delegate
Known location and testable symptom; tests, docs, tech debt

Keep human-led
Broad refactors, tacit domain logic, security-sensitive work

Plan first
Larger work: plan in the Agents tab, then ask for a PR

✅ What the PR owner checks first

- **Plan matches the issue's intent?**
- **Which commit holds the code?**
- **Changes outside expected paths?**
- **Tests added, and has anyone run them?**
- **Runs awaiting approval? Evidence absent?**

🔒 Platform-fixed: drafts on a copilot/ branch only; a person other than the requester approves and merges.

PART 2

Reason About Intent

Read which commit each review analyzed, run what it left unexecuted, and choose who approves



Which head?

The automatic review saw the plan-only commit



Run the tests

Review named them unexecuted; the maintainer ran 8



#198 Retry

Review found an inert button four tests missed

```
re-requested Copilot review on 02ddaf3
↳ 🟢 approval recommended · tests not executed → maintainer runs them
```

The Automatic Review Analyzed the Plan-Only Head

FanHub ruleset: automatic review on every PR to main, including drafts · review_on_push: false

❌ Automatic review (ruleset)

1. **Fires at 20:31:37**
   Six seconds after the draft opens
2. **Analyzes 919b158**
   The Initial plan commit, before any code
3. **“Wasn't able to review any files”**
   A plan-only head has no code to read
4. **Fix lands at 20:34:55**
   review_on_push: false, so no new review

📄 A review of the plan

✅ Re-requested review (maintainer)

1. **Maintainer re-requests Copilot**
   From the Reviewers sidebar
2. **Analyzes 02ddaf3**
   The fix plus 8 regression tests
3. **Approval recommended**
   Tests named as not executed
4. **Assessment names its commit**
   The reviewer knows what it covers

🔎 A review of the implementation

Turn on Review new pushes, or read the first automatic review as a review of the plan

Copilot Recommended Approval; the Maintainer Ran the Tests

maintainer terminal · #210 head 02ddaf3

```
$ CI=true npx react-scripts test --watchAll=false \
    src/pages/Episodes.test.js
PASS src/pages/Episodes.test.js
  ✓ filtered responses do not overwrite the …
  ✓ cache hits restore season controls on …
  ✓ a late filtered response cannot replace …
  ✓ an unmounted request cannot overwrite …
   … 4 more
Tests: 8 passed, 8 total
```

🧠 Copilot code review: assessment

"🟢 Approval recommended … regression tests cover the affected behavior but were not executed."

🧪 Maintainer: executed evidence

Runner status PASS, 8 of 8. Domain result: filters, expiry, remounts, and races behave as #7 expects

📜 Two answers, two questions

Review judged the diff sound and named the step it left for a person

💡 Observed on FanHub #210. A green assessment and an executed run each answer a different question.

On #198, Review Caught an Inert Retry That Tests Missed

❌ Green build, inert button

4 component tests passed
Build green on FanHub draft #198

Static render
The Home page had no interactive render mode, so Retry never wired up

💡 Copilot review found

No interactivity
The bUnit click test could not see the browser boundary

Alert cleared too early
The Retrying... state was unreachable

```
@rendermode InteractiveServer
```

✅ The author verified

A new render-mode test failed against the original draft

6 of 6 tests pass after the fix

Browser against an API returning 503: alert shows, Retry restores the page

🔎 Observed on FanHub #198. Review reads intent, and intent is where component tests often stop.

Hand Fixes Back; Decide Whether Copilot's Approval Counts

🔁 Fix with Copilot · public preview

Comment → cloud agent
Copilot pushes a commit or opens a new pull request

Shared setup
Review reads the same instruction files and MCP configuration as the cloud agent

PR owner stays in the loop
Each pushed commit is new evidence to check

⚖️ Copilot approvals · public preview

Admin opt-in (Sept 2026)
Every review has an approval assessment; opted-in approvals count

Dismissed on new pushes
The same rule as a human approval

Scope it
Keep a human approval where behavior, security, or data is at stake

🔑 Counting Copilot's approval promotes advice to authority. Choose the paths, the owner, and the way back.

PART 3

Measure Against Rules and Thresholds

Use CodeQL findings, ratings, and coverage for what rules measure; tests and reviewers cover behavior

📏

Standard findings

CodeQL rules, reproducible, can gate (GA)

✨

AI findings

Model-mediated, default branch, advisory (preview)

🚦

#199 coverage gate

Blocked at 26.7%, passed at 30% (observed)

```
Code Quality run on #210 head 02ddaf3
↳ 0 findings · CodeQL reported no new alerts (observed)
```

Standard Findings Can Gate a Merge; AI Findings Advise

🏷️ Standard findings · GA

Source
CodeQL quality rules

Reproducibility
Stable for the same code, rules, and tool version

Where
Pull requests and default branch, with Autofix suggestions

Gate
Can block merge by severity threshold

✨ AI findings · public preview

Source
Model review of recently changed files

Reproducibility
Model-mediated

Where
Default branch only

Gate
Advisory; cannot block merge

🧪 #210 (observed): the run on 02ddaf3 posted no findings. Rules cover known patterns; the cache behavior still needs executed tests and a reviewer.

On #199, a Coverage Rule Blocked, Then Passed

Active ruleset: Cobertura upload required, 27% minimum on the target branch (observed)

30%

coverage on draft #199 after two tests, up from a blocked 26.7%

Observed: FanHub #199

🚫 **First head blocked**

6 tests passed; 26.7% fell under the 27% rule

✍️ **Author added two tests**

Hand-written Home page tests; 8 of 8 pass

✅ **Same PR turned green**

GitHub displayed 30% and the gate passed

🤖 **Findings → cloud agent (GA)**

Comment @Copilot on a finding or bulk-assign from the dashboard

🚦 Start each new rule in Evaluate mode to see what it would block, then promote it to Active once the team trusts the signal.

PART 4

Decide What Gets Authority

Map evidence to the people who rely on it, then permit, count, or enforce one setting at a time

🗺️

Five surfaces

Context, invocation, review, merge evidence, cost

🪜

An uneven ladder

Permit the producer, count review, enforce rules

📝

#210 decision record

Written before acting, with a recovery path

```
platform owner promotes one setting
↳ named owner + observation period + rollback condition
```

Each Agent Hands the Decider One Kind of Evidence

#210 carried all three on 02ddaf3; the maintainer added an executed run before deciding

Cloud agent	Draft diff, own tests, linked issue; people judge scope and proof	`produces`
Code review	Comments and an approval assessment; people weigh each and run tests	`reasons`
Code Quality	Findings, ratings, coverage, Autofix; security review stays separate	`measures`
Maintainer	Executed check (8 of 8 on #210), residual risk, and the merge	`decides`

Each agent supplies its own kind of evidence, and only that kind

Five Surfaces Shape What the Three Agents Do

The platform owner works across all five

Agent context	copilot-instructions.md, *.instructions.md, AGENTS.md, custom agents	`agent + review`
Invocation	copilot-setup-steps.yml, firewall, MCP config, workflow approvals	`cloud agent`
Review policy	Rulesets: automatic review, Review new pushes, Copilot approvals	`code review`
Merge gates	Code Quality settings, CodeQL, coverage upload, quality rulesets	`Code Quality`
Economics	AI Credits, Actions minutes, Code Quality licenses	`every promotion`

Shared context: excludeAgent: code-review keeps a test-writing rule with the producer and off the reviewer's checklist

Promote Advice to Authority One Step at a Time



Cloud agent · permit

Observe
Assign a few scoped issues; read every draft

Evaluate
Add copilot-setup-steps and custom agents

Enforce
Platform-fixed: drafts only; a person approves and merges



Code review · count

Observe
Request reviews manually

Evaluate
Ruleset auto-review with Review new pushes

Enforce
Count Copilot approvals (preview) on chosen paths



Code Quality · enforce

Observe
Enable on one repository; read the backlog

Evaluate
Quality and coverage rules in Evaluate mode

Enforce
Active rules at an accepted threshold; AI findings stay advisory

↩ Each promotion has an owner and a way back: revert to Evaluate, dismiss with a reason, or close and reassign. Budget AI Credits and $10 per committer for Code Quality.

The Decision on #210, Written Before Acting

The maintainer's record for head 02ddaf3

Available	Patch + tests, approval assessment, clean quality run, 8/8 executed	`on the head`
Absent	A browser check of the season filter; a Node frontend coverage gate	`named gap`
Policy met	Default-branch ruleset; CodeQL with no new alerts	`rules`
Judgment	One page and its tests; the tests exercise the reported symptom	`person`
Action	Mark ready and squash-merge; recovery: revert and reopen #7	`action + recovery`

Observed: squash-merged as 803254f by the maintainer; #7 closed. Recovery stays ready: revert 803254f and reopen #7

From One Green Check to Evidence With a Named Producer

Before

- Evidence read as one combined signal
- Review results read without their commit
- Tests the review named left for later
- Authority granted to agents as a bundle

After

- Each item traced to its producer and its head
- Review new pushes on, or the first review read as a plan review
- A person runs the tests the review names
- One promoted setting with an owner and a rollback condition

12 s
Assignment to draft #210 (observed, one run)

8 of 8
Tests passed when the maintainer ran them

26.7% → 30%
#199 coverage gate blocked, then passed

✅ WHAT YOU CAN DO TODAY

Today

- Assign one bug with a known location and testable symptom to Copilot; time the draft
- Check your review ruleset for Review new pushes
- Read which commit each Copilot review analyzed

This Week

- Run the agent's tests yourself before trusting a green assessment
- Add one excludeAgent rule to an instruction file and watch one agent change
- Write a decision record for your next agent-authored pull request

This Month

- Enable Code Quality on one repository with one coverage rule in Evaluate mode
- List the evidence on your last merged PR's final head and name the missing kind
- Promote one setting with a named owner and a rollback condition

🔑 Key Takeaway

Produce, reason, measure, then a person decides, with every promotion owned and reversible.

REFERENCES

Cloud Agent and Code Review

About Copilot cloud agent
Runtime, firewall, and authority

Configure automatic code review
Rulesets and Review new pushes

excludeAgent instructions
Agent-specific context

Approvals and Code Quality

Copilot approvals (preview)
Approvals that count

About GitHub Code Quality
Findings, ratings, rules

Code Quality billing
Per-committer pricing

FanHub Evidence

Draft #210
Cloud agent fix for #7

#198
Review finds an inert Retry

#199
Coverage gate blocks, then passes

Where It Stops Fitting

GitHub Agentic Workflows
Recurring judgment on schedules and events

Agents at Your Keyboard
Local plan, implement, verify loops

Agentic SDLC
Readiness score and maturity path



Copilot on GitHub

PRODUCE, REASON, MEASURE. THEN A PERSON DECIDES.

Which head?

The automatic review analyzed the plan; re-request or enable Review new pushes

8 of 8

Copilot named the tests as unexecuted; the maintainer ran them

26.7% → 30%

A coverage rule blocked, then passed, on #199

Take the last PR your team merged: which kinds of evidence sat on its final head, and which single setting would you promote first?